

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 11: Time Systems, Solar Phenomena & Twilight

Syllabus Source: CAE Oxford (pp. 409-452) | Pass Target: $\geq 90\%$ AviationExam

1. EARTH ROTATION & ARC-TO-TIME CONVERSION

The Earth rotates 360° of longitude in 24 hours (1 mean solar day):

- Arc to Time Rates: $360^\circ = 24$ hours; $15^\circ = 1$ hour (60 min); $1^\circ = 4$ minutes; $1'$ (minute of arc) = 4 seconds.
- Coordinated Universal Time (UTC / Z): The atomic time standard referenced to the Greenwich Meridian (000° Longitude).
- Local Mean Time (LMT): The solar time at a specific meridian of longitude: $LMT = UTC \pm$ (Longitude in time). Memory rule: Longitude EAST is LATER (ADD to UTC); Longitude WEST is EARLIER (SUBTRACT from UTC) ('East Ahead, West Behind').
- International Date Line (IDL): Located along the 180° meridian. Crossing WESTBOUND into East longitude: ADVANCE ONE CALENDAR DAY (+1 day). Crossing EASTBOUND into West longitude: GO BACK ONE CALENDAR DAY (-1 day)!

2. CIVIL TWILIGHT DEFINITION

Civil Twilight is defined as the interval of time between sunset and the moment the CENTER OF THE SUN IS 6° BELOW THE CELESTIAL HORIZON (in the morning, from 6° below horizon to sunrise). It marks the legal limit for VFR day operations!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Arc to Time Conversion & UTC to

Arc-to-Time Table: - $360^\circ = 24$ hours | $15^\circ = 1$ hour | $1^\circ = 4$ minutes | $1'$ of arc = 4 seconds. Problem: UTC is 12:00. What is Local Mean Time (LMT) at $075^\circ 15' W$? Step 1: Degrees: $75^\circ \times 4 \text{ min} = 300 \text{ min} = 5$ hours. Step 2: Minutes of arc: $15' \times 4 \text{ sec} = 60 \text{ sec} = 1$ minute. Step 3: Total time difference = 5 hours 01 minute. Step 4: West is earlier (subtract): $LMT = 12:00 - 05:01 = 06:59$ LMT!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Converting UTC to Local Mean Time (LMT)

SCENARIO (Standard EASA Solar Time Drill): At position $40^\circ N$ $075^\circ W$, a pilot wants to know the Local Mean Time (LMT) when it is 16:30 UTC: QUESTION: What is the exact Local Mean Time (LMT)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify Longitude and Hemisphere: - Longitude = 075° WEST. Step 2: Convert Longitude (degrees) into Time: - Rate: 15° of longitude = 1 hour of time. - Time difference = $75^\circ / 15^\circ = 5.0$ hours (5 hours 00 minutes)! Step 3: Apply the East/West Rule to UTC: - West Longitude is BEHIND Greenwich time ('West is Behind' -> SUBTRACT!). - $LMT = UTC -$ Time Difference - $LMT = 16:30 - 05:00 = 11:30$ LMT! FINAL ANSWER: Local Mean Time at $075^\circ W$ is exactly 11:30 LMT (the sun is nearly overhead at local noon).

[!] EXAM TRAP (AVIATIONEXAM) - The Civil Twilight 6° Below Horizon Trap

- Civil Twilight ends when the sun center is 6° BELOW the horizon (legal daylight ends). - Nautical Twilight is 12° below horizon. - Astronomical Twilight is 18° below horizon. - Aviation law uses CIVIL TWILIGHT (6°) exclusively!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: If UTC is 12:00, what is the Local Mean Time (LMT) at longitude 045°E?**

- [A]: 09:00
- [B]: 15:00 LMT ($45^\circ / 15^\circ = 3$ hours East -> ADD 3 hours to 12:00)
- [C]: 12:45
- [D]: 18:00

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Longitude 45° East is 3 hours ahead of Greenwich ($45 / 15 = 3$). LMT = 12:00 + 3h = 15:00.

Q2: In aviation regulations, when does EVENING CIVIL TWILIGHT officially end?

- [A]: At sunset
- [B]: When the center of the sun's disc is 6 DEGREES BELOW the true horizon
- [C]: 30 minutes after sunset
- [D]: At complete darkness

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Civil twilight is defined astronomically as the period ending when the sun's center reaches 6° below the sensible horizon.

Q3: An aircraft flies across the International Date Line (180° meridian) from WEST TO EAST (e.g. from Tokyo to Sydney).

- [A]: Advance one day
- [B]: SUBTRACT ONE DAY (repeat the same calendar date)
- [C]: Zero change
- [D]: Advance 12 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Crossing eastbound from East longitude to West longitude across the 180° line repeats the current day (subtract 1 day).